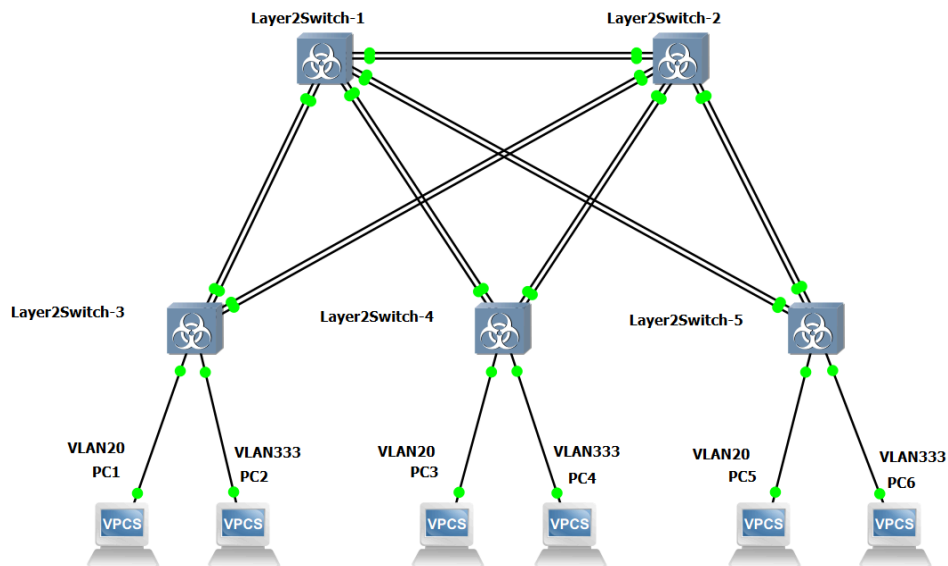


Тема: Настройка виртуальной локальной сети (VLAN)

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN

Скопировал шаблон lab3.



Настройка адресов для VPCS.

```
PC1> ip 192.168.20.11
Checking for duplicate address...
PC1 : 192.168.20.11 255.255.255.0

PC2> ip 192.168.33.12
Checking for duplicate address...
^CPC2 : 192.168.33.12 255.255.255.0

PC3> ip 192.168.20.13/24
Checking for duplicate address...
PC3 : 192.168.20.13 255.255.255.0

PC4> ip 192.168.33.14/24
Checking for duplicate address...
PC4 : 192.168.33.14 255.255.255.0

PC5> ip 192.168.20.15/24
Checking for duplicate address...
PC5 : 192.168.20.15 255.255.255.0

PC6> ip 192.168.33.16
Checking for duplicate address...
```

PC6 : 192.168.33.16 255.255.255.0

К VLAN 20 принадлежат PC1, PC3, PC5. К VLAN 333 принадлежат PC2, PC4, PC6.

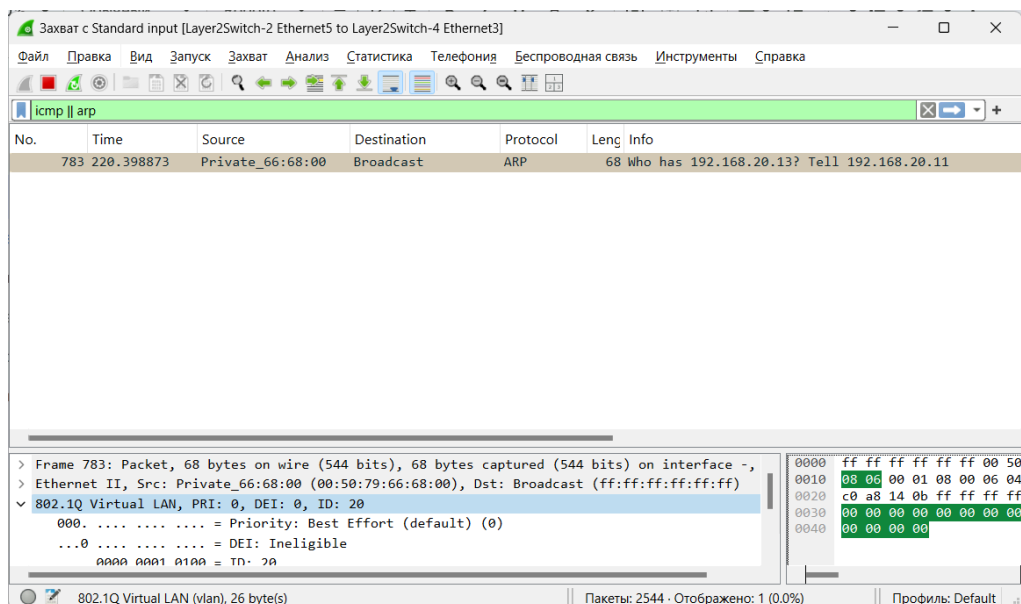
На коммутаторах установил порты для PC в режим access, остальное trunk, установил trunk encapsulation dot1q, (было указано isl, теги не добавлялись). Указал vlan 333 как native для магистральных портов. Конфигурации коммутаторов находятся в файлах.

```
PC1> ping 192.168.20.13
```

```
84 bytes from 192.168.20.13 icmp_seq=1 ttl=64 time=2.838 ms
84 bytes from 192.168.20.13 icmp_seq=2 ttl=64 time=2.183 ms
84 bytes from 192.168.20.13 icmp_seq=3 ttl=64 time=20.681 ms
84 bytes from 192.168.20.13 icmp_seq=4 ttl=64 time=2.366 ms
84 bytes from 192.168.20.13 icmp_seq=5 ttl=64 time=14.339 ms
```

После настройки режимов портов на коммутаторах 3, 4 и 5, при попытке отправить ping, начинался шторм из ARP-запросов и ping тормозил или вообще возвращал timeout. Это странно, так как show spanning-tree на коммутаторах показывал, что STP работает. В Wireshark проверил и обнаружил, что из всех запросов только первый содержал заголовок "802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20". Собственно, оказалось что Layer2Switch-2 не добавляет поле тега, потому что у него было указано поле encapsulation isl, а должно быть dot1q, я его пропустил при настройке и он из-за этого хаотично пересылал нетегированный трафик.

После исправления проверил на нескольких линках захват, лишнего не передается.



2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать

```
PC1> ping 192.168.20.13

84 bytes from 192.168.20.13 icmp_seq=1 ttl=64 time=2.838 ms
84 bytes from 192.168.20.13 icmp_seq=2 ttl=64 time=2.183 ms
84 bytes from 192.168.20.13 icmp_seq=3 ttl=64 time=20.681 ms
84 bytes from 192.168.20.13 icmp_seq=4 ttl=64 time=2.366 ms
84 bytes from 192.168.20.13 icmp_seq=5 ttl=64 time=14.339 ms

PC1>
PC1> ping 192.168.20.13

84 bytes from 192.168.20.13 icmp_seq=1 ttl=64 time=7.623 ms
84 bytes from 192.168.20.13 icmp_seq=2 ttl=64 time=9.684 ms
84 bytes from 192.168.20.13 icmp_seq=3 ttl=64 time=2.457 ms
84 bytes from 192.168.20.13 icmp_seq=4 ttl=64 time=8.535 ms
84 bytes from 192.168.20.13 icmp_seq=5 ttl=64 time=5.004 ms

PC1> ping 192.168.20.15

84 bytes from 192.168.20.15 icmp_seq=1 ttl=64 time=11.223 ms
84 bytes from 192.168.20.15 icmp_seq=2 ttl=64 time=8.272 ms
84 bytes from 192.168.20.15 icmp_seq=3 ttl=64 time=14.325 ms
84 bytes from 192.168.20.15 icmp_seq=4 ttl=64 time=2.358 ms
84 bytes from 192.168.20.15 icmp_seq=5 ttl=64 time=16.858 ms

PC1> ping 192.168.33.16

No gateway found

PC1> ping 192.168.33.12

No gateway found
```

Тут ping не проходит, потому что это сеть другая, но если поменять IP какого-нибудь хоста из VLAN333, то запрос все равно не пройдет:

```
PC6> ip 192.168.20.16
Checking for duplicate address...
PC6 : 192.168.20.16 255.255.255.0

PC1> ping 192.168.20.16

host (192.168.20.16) not reachable
```

После перезапуска PC6, восстановится изначальный адрес, теперь попробуем отправить запрос с него:

```
Checking for duplicate address...
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

```
PC6 : 192.168.33.16 255.255.255.0

PC6> ping 192.168.20.15

No gateway found

PC6> ping 192.168.33.14

84 bytes from 192.168.33.14 icmp_seq=1 ttl=64 time=2.842 ms
84 bytes from 192.168.33.14 icmp_seq=2 ttl=64 time=10.343 ms
84 bytes from 192.168.33.14 icmp_seq=3 ttl=64 time=5.256 ms
84 bytes from 192.168.33.14 icmp_seq=4 ttl=64 time=16.055 ms
84 bytes from 192.168.33.14 icmp_seq=5 ttl=64 time=4.034 ms
```

Хосты из VLAN20 доступны только в VLAN20, а хосты из VLAN333 только в VLAN333.

3) Перехватить в Wireshark пакеты с тегами и без тегов (nb!), результаты задокументировать

No.	Time	Source	Destination	Protocol	Leng	Info
783	220.398873	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.13? Tell 192.168.20.11
4382	1168.836494	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.13? Tell 192.168.20.11
4420	1177.634772	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.15? Tell 192.168.20.11
4448	1184.255218	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
4456	1185.256810	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
4457	1186.255901	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11

```
> Frame 783: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 000000000000000000000000000000000000000000000000
    Trailer: 00000000
> Address Resolution Protocol (request)
```

```
> Frame 783: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 0000000000000000000000000000000000000000
    Trailer: 00000000
> Address Resolution Protocol (request)
```

[illegible]

```
> Frame 783: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 0000000000000000000000000000000000000000
    Trailer: 00000000
> Address Resolution Protocol (request)
```

[illegible][illegible]

```
> Frame 783: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .. = Priority: Best Effort (default) (0)
    ...0 .... .. = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
Type: ARP (0x0806)
Padding: 0000000000000000000000000000000000000000000000000000000
Trailer: 00000000
> Address Resolution Protocol (request)
```

[illegible][illegible]

```
> Frame 783: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
v 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 0000000000000000000000000000000000000000
    Trailer: 00000000
> Address Resolution Protocol (request)
```

No.	Time	Source	Destination	Protocol	Leng	Info
783	220.398873	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.13? Tell 192.168.20.11
4382	1168.836494	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.13? Tell 192.168.20.11
4420	1177.634772	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.15? Tell 192.168.20.11
4448	1184.255218	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
4456	1185.256810	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11

[illegible][illegible]

```
> Frame 4420: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
Type: ARP (0x0806)
Padding: 00000000000000000000000000000000000000000000
Trailer: 00000000
> Address Resolution Protocol (request)
```

[illegible][illegible]

```
> Frame 4420: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
√ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 ..... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
Type: ARP (0x0806)
Padding: 00000000000000000000000000000000
Trailer: 00000000
> Address Resolution Protocol (request)
```

```
> Frame 4420: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
√ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 00000000000000000000000000000000
    Trailer: 00000000
> Address Resolution Protocol (request)
```

[illegible][illegible]

```
> Frame 4420: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
√ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
    000. .... = Priority: Best Effort (default) (0)
    ...0 .... = DEI: Ineligible
    .... 0000 0001 0100 = ID: 20
    Type: ARP (0x0806)
    Padding: 00000000000000000000000000000000
    Trailer: 00000000
> Address Resolution Protocol (request)
```

С тегами идут фреймы отправленные через access порты, которые добавляют тег vlan 20. Тут запросы от 192.168.20.13 и 192.168.20.15 (vlan 20).

А без тега идут фреймы от хостов в vlan 333, так как vlan 333 указан как native для магистральных портов.

icmp arp						
No.	Time	Source	Destination	Protocol	Leng	Info
5134	1359.759757	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
5165	1368.514922	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
5167	1369.515590	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
5176	1370.516771	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.16? Tell 192.168.20.11
5376	1423.747639	Private_66:68:05	Broadcast	ARP	64	Gratuitous ARP for 192.168.33.16 (Request)
5385	1424.753309	Private_66:68:05	Broadcast	ARP	64	Gratuitous ARP for 192.168.33.16 (Request)
5386	1425.748165	Private_66:68:05	Broadcast	ARP	64	Gratuitous ARP for 192.168.33.16 (Request)
5560	1471.571254	Private_66:68:05	Broadcast	ARP	64	Who has 192.168.33.14? Tell 192.168.33.16

> Frame 5560: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

▼ Ethernet II, Src: Private_66:68:05 (00:50:79:66:68:05), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Destination: Broadcast (ff:ff:ff:ff:ff:ff)

> Source: Private_66:68:05 (00:50:79:66:68:05)

Type: ARP (0x0806)

[Stream index: 26]

Padding: 00

Frame check sequence: 0x00000000 [unverified]

[FCS Status: Unverified]

> Address Resolution Protocol (request)

4) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

Сохранены.